

AetherDB Security Policy & Hardening Guide

1. Network Security and Ports

AetherDB nodes expose three key ports: 8080 (REST Client API), 9090 (Raft Consensus), and 7070 (Gossip Protocol). Firewalls must be configured to prevent unauthorized external access. Specifically, Raft Consensus port 9090 and Gossip port 7070 must only be reachable by cluster members.

2. Authentication and Authorization

All data manipulation requests must be authenticated. The X-Aether-Token HTTP header is mandatory for accessing `/api/v1/write` and `/api/v1/read`. Health check requests on `/api/v1/health` do not require authentication.

Tokens should be cryptographically random and rotated every 30 days. Shared tokens across clients are strictly prohibited.

3. Encryption at Rest

To protect stored data from physical theft, encryption at rest must be enabled. AetherDB supports AES-256 block encryption for the WAL and SSTables. The encryption key must be managed using an external Key Management Service (KMS).